

Dear Diary, I won't give up my CAVE!

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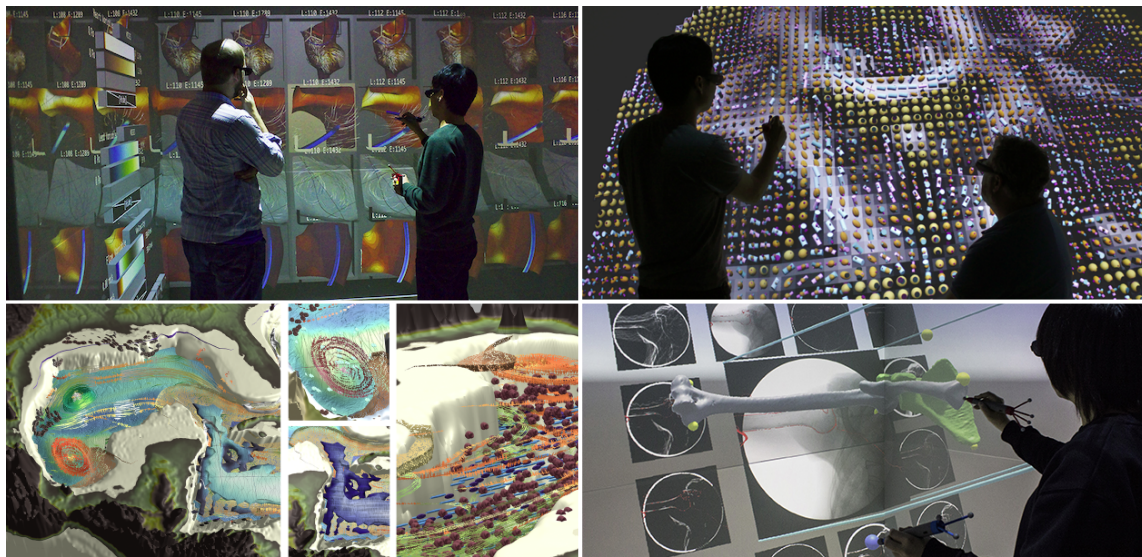


Figure 1: A CAVE has often been the centerpiece of our Immersive Visualization Laboratory.

ABSTRACT

Daniel F. Keefe is of Professor of Computer Science and Engineering and Director of the Interactive Visualization Lab at the University of Minnesota. Together with other visual computing faculty and students, Keefe's lab recently moved to a custom designed, newly renovated floor of a research building on campus. That is important for this position paper because the move raised the key question: *What will happen to the CAVE?* After 25 years of working with various large format semi-immersive multi-wall projection screen displays that are both loved and hated for their high-performance visualization capabilities and challenges—the author reflects on the strategy, finances, emotions, interdisciplinary impacts, and technical considerations that have impacted his decision to keep his CAVE alive despite the rapidly evolving landscape for immersive computing. To emphasize the multi-faceted nature of the decision, including the emotion we humans can experience as users and developers of a technology like this, which is almost unique in its ability to both define a physical architectural space and a wide range of complementary virtual spaces, the position paper takes the creative form of a series of entries in a personal diary.

Index Terms: CAVE, VR, immersive visualization lab.

1 PLANNING - SPRING 2023 TO 2025

Dear Diary,

I need your help, and I don't know what to do. I'm absolutely thrilled that our visual computing program (lab spaces and faculty

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Figure 2: The reconstructed and partially upgraded (work still in progress) CAVE installed in the new visual computing labs on the 3rd floor of the Shepherd Labs building at the University of Minnesota, Minneapolis, USA.

and student offices) will be moving to the third floor of the newly renovated Shepherd Labs building next year, but now the architects need some answers about what the space should look like, and we need to make some important decisions that will hopefully help us to succeed for years to come.

The question I need to answer right now is: *What will happen to the CAVE?*¹ Is this technology part of the future of our visual

¹Our CAVE is a 4-wall version inspired by the original created by and

computing program at the University of Minnesota or is it a relic of the past? The context, of course, is that VR, AR, MR, and XR (I'll just use VR as a shorthand for all these) have changed so much with the introduction of commodity off-the-shelf (COTS) head-worn displays. I'm now able to teach my classes with a fleet of VR devices that students simply check out for the semester. From art classes to computer science classes to high-end data visualization, we use COTS VR hardware for all of these today, and this has opened up VR for so many more people than before. Will the CAVE still be useful in this new context?

The pros and cons I see right now are as follows. On the positive side:

- The centerpiece argument: A CAVE has always been a centerpiece of our lab (Figure 1). A large display like this makes an exciting architectural feature for the new space. It is a centerpiece and a place to gather.
- The demo argument: We have always been able to show exciting work in the CAVE that impresses visitors in government, industry, and academia.
- The collaboration argument: I don't believe it when people say it works just as well to collaborate on data visualization while wearing headsets. In 25 years of practice, the CAVE has always been awesome for this, and this is a key to so many of our projects.
- One view of the accessibility argument: With a CAVE, we have always been able to include everyone who comes to our lab—one might say that it degrades gracefully. Users can wear glasses or not, they can sit or stand, and they don't have to mess up their hair or feel self-conscious putting on a device. We are working with a wide range of collaborators now, from climate scientists and doctors to kids and elders from local Indigenous communities. Head-worn displays simply do not work for everyone.

On the negative side:

- The maintainability argument: This is a custom-designed instrument that we designed and built ourselves (Figure 2), I'm not even sure if we can successfully move the structure and reassemble it in a new building.
- The VR technologies have moved on argument: The COTS devices control the market right now. It's difficult to find projectors, glasses, and other hardware. Some game engines have dropped support for quad-buffered stereo.
- The other side of the accessibility argument: While we do very well making the CAVE in our lab accessible to lots of visitors and collaborators, and they tend to prefer it to head-worn displays, it is only as accessible as our lab. I would never recommend that our community partners or even our collaborators from other disciplines to install a CAVE like ours without a plan to hire someone to run it and maintain it—it's like a planetarium—very accessible if you go to it but not so accessible at your home or office.

Notice, that I did not directly include a positive or negative argument on the scientific merits. That's because I am struggling a bit with where I land on that, and this is particularly difficult because in my role as a computer scientist researcher I would typically think the science *should be* the most important argument to me. The key question for this argument is this:

registered as a trademark of the University of Illinois at Chicago's Electronic Visualization Laboratory [1].



Figure 3: The pieces of the square frames that the projection screens for the walls are stretched across did not survive the move. To save money, I fabricated these 12 replacement pieces over several weekends in my garage.

Is there a research question about how humans will interact with computers and data in the future that we can only answer with the help of a technology like the CAVE?

My thinking right now is that all of the investment that has gone into the COTS devices makes them a better fit for answering the traditional research questions that we ask in VIS and VR research, for example, questions dealing with high-resolution accurate spatial perception or presence and immersion. So, maybe the CAVE simply can't compete for these more traditional questions.

However, I think the CAVE may still be a unique research tool to enable study of larger interdisciplinary questions that go beyond traditional computer or perceptual science. For example, I would like to work with social scientists, humanists, and design researchers to answer questions like: How will we work with data and computing in a future where we regularly redefine and build both the physical and virtual spaces that we work in? As a precursor for this, how does the physical cube of the CAVE impact our work and how do we effectively engage with this physicality rather than try to make the walls disappear? What physical objects and activities does it make sense to bring into interactive hybrid digital+physical spaces like the CAVE? These questions are super exciting and relevant but they require thinking a bit broader about the type of research we conduct with a CAVE.

So, I think we'll move it. I just hope it doesn't break!



Figure 4: Replacement projection screens arrive in shambles days before the big opening demonstration and ribbon cutting ceremony with the College of Science and Engineering Dean's Advisory Board.

2 MOVING IN LATE SUMMER / EARLY FALL 2025

Dear Diary the week before moving,

I'm back. We are getting ready to move the lab to the new space. Unfortunately, I injured myself this weekend, and now I am not supposed to lift anything too heavy. We will have professional movers on the day of the move, but their official line is that "anything fragile we should move ourselves". They are certainly not going to be able to disassemble and reassemble the CAVE. So, I have decided to hire my son and two other members of his high school robotics team to help my students. Together with my grad students, their robotics background and good teamwork should be just what is needed for this project! Is this how research is supposed to work?

Dear Diary one day before moving,

The team did a marvelous job disassembling and packing up the CAVE. We did have one hardware casualty. The frames that the projection screen walls are stretched across are made of clear acrylic, and it became brittle over the years. So, several pieces cracked during the disassembly. They should really be made of something stronger like polycarbonate.

Dear Diary on moving day,

The moving team got all the parts into the new space *except* for the floor, which is an 8 foot by 8 foot square that does not fit into the elevator of the new building. Ugh.

Dear Diary the weekend after moving,

So, it turns out that polycarbonate is quite expensive, and it would cost thousands of dollars to fabricate those framing pieces that broke to a more robust specification. I do have the money to pay for it, but it would leave less money to upgrade the projectors, graphics cards, and computer; which I know will be incredibly important. So, I am going into do-it-yourself wood working mode to fabricate replacement frames in my garage and also to assemble a new floor from panels that are small enough to fit into the elevator. (See Figure 3 for photographic evidence of garage woodworking progress.) Is this part of the job description for computer scientists?

Dear Diary during the first month of the semester,

I'm sorry for not writing earlier. I have been too busy doing woodworking with a router, track saw, and drill press in my garage. I have learned so much. I think I should now be able to teach this as a second job! I have also been trying to find the best way to build a raised wooden floor for the CAVE on top of the lab's flooring, which I am discovering is not at all level!

Dear Diary the week before the grand opening,

Today is Thursday. It turns out the company lost the purchase request for the replacement projection screen material that I ordered in July. So, the shipment has been months late. This hasn't been a problem until now because I have been so busy with my wood-working, but now we need to get the screens installed as soon as possible so that we can do a demo for the ribbon cutting ceremony that the Dean of the College of Science and Engineering is hosting on Monday together with his advisory board of important alumni and industry partners. At least the tracking number says the package will arrive Thursday. Then, we can work through the weekend to get the CAVE running.

Dear Diary Thursday morning,

The projection screen arrived, and this is a complete disaster! (See Figure 4 for photographic evidence.) It should have arrived rolled on a large cardboard tube to protect it, but instead it was folded up in a box with no padding. This is completely unusable.

Dear Diary Thursday afternoon,

I just got off the phone with the projection screen company. They feel terrible about this and are going to overnight a replacement to deliver to my house by tomorrow (Friday) evening!

Dear Diary Monday afternoon,

The projection screen arrived Friday. I brought food and coffee, and my students and I worked through the weekend to hang up the projection screen, align the projectors, reconnect the computer, and get a small demo finally working about one hour before the ribbon cutting ceremony. Luckily, nobody will ever know how close we came to disaster!

3 REFLECTIONS IN WINTER 2025/2026

Dear Diary,

I still haven't finished upgrading the CAVE because I am having trouble deciding which projectors would provide a worthwhile upgrade. It is very difficult to find the right thing, and this is a clear illustration of the "maintainability argument" against systems like this.

With this said, I have also had some time to reflect on the adventure of the past few months of moving this research instrument, and I would like to share two lessons learned.

The first lesson is that I am willing to admit that I have an emotional attachment to this special research instrument called the CAVE. 25 years ago, I published the first research paper of my career on the CavePainting system, and it had Cave right in the title. In 2014, my research community at the IEEE VR conference visited



Figure 5: Okay, I admit it. My emotional attachments to the CAVE include. Left: The first research paper I published, literally titled “CavePainting” [2]. Middle: The centerpiece of lab tours and demonstrations at the IEEE VR 2014 conference held in Minneapolis. Right: A professional photograph to accompany the announcement of a teaching award was not posed in front of the classroom as you might expect but instead right in front of the CAVE.

my lab to see our work in the CAVE. When I have been recognized by the University with an award, even one unrelated to our research, what is it that the photographer wants to shoot and that I am proud to show together with my students? Of course, it is our software running in our Cave. (See Figure 5.) So, I will just go ahead and admit that this is a special emotional attachment. But, why? Can I learn anything from this attachment?

My current theory is that it all comes down to the physical presence of the CAVE. It is like a work of art or architecture or like a favorite wooden canoe. It has a presence in the physical spaces of our labs; yet, it can also take us anywhere through a virtual environment. This device is special precisely because of its physical presence. On one hand, that makes it much harder to move than a head-worn display, but on the other, I would never form an emotional attachment to a commodity head-worn display. I’m not saying everyone has to go build a CAVE and fall in love with it—I’m just saying, try thinking about this computer device the way an artist or designer, like an architect, would think about it—the scale and physical presence of this digital device is precisely what makes it special. Thus, for a CAVE, maybe the special feature of the technology is not so much that it can create a level of immersion that leads to an illusion of presence—that can be done even better with other devices. Perhaps the special feature is just the opposite—when we enter the CAVE, we can travel into a virtual environment *while also knowing* that we continue to exist within the physical architectural space of the device itself.

The second lesson is a reflection not just on the CAVE but on the values this technology can represent in the particular time that we live in. I’m speaking of two main trends. First, we are now just a few years after a global pandemic that dramatically changed how we work and interact with each other, many people say “forever”. The main shift is a dramatic increase in remote interactions as well as asynchronous interactions supported by technology. While there are clear benefits for scalability, efficiency, and accessibility; there are also drawbacks, particularly in terms of quality deep communication and engagement with each other. Just think of how much multitasking goes on during today’s online meetings. I don’t think we yet fully understand the implications. Second, the rapid growth of AI-based technologies seems to be having a compounding effect on this trend, automating and accelerating many of our interactions with technologies and each other but often with questionable quality. In short, it is easier to communicate but the social context and technologies makes it more difficult to communicate as deeply.

In this context, when I think of what I want to research as a computer scientist, I feel called to research and invent technologies that are designed specifically to bring people together, even physically. I

want my lab to demonstrate the benefits of physically bringing students and interdisciplinary collaborators together. In other words Diary, I won’t give up my CAVE!

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